

Instructions to the Students:

1. Solve ANY FIVE questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. All sketches should be labeled, neat and proportionate.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

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|--|--------|---|
| A) State the comparative merits and demerits of stone masonry and brick masonry? | (CO 1) | 6 |
| B) Draw neat sketches, showing plan of two consecutive courses of 1½ brick thick corner of wall in English bond? | (CO 1) | 6 |
| C) What are the general principles observed in brick masonry? | (CO 1) | 6 |

Q.2 Solve Any Two of the following.

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|--|--------|---|
| A) What are the chief ingredients of plain cement concrete? Describe in detail the properties of each? | (CO 2) | 6 |
| B) Describe the properties of fresh and hardened concrete? | (CO 2) | 6 |
| C) Explain the role of admixtures in concrete? | (CO 2) | 6 |

Q.3 Solve Any One of the following.

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|--|--------|----|
| A) Sketch a semicircular arch showing all technical terms used in arch? Sketch the entire arrangement showing the details of centering work? | (CO 3) | 12 |
| B) Explain Brick lintel, reinforced concrete lintel and steel lintel with sketches? | (CO 3) | 12 |

Q.4 Solve Any Two of the following.

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|--|--------|---|
| A) Draw a neat sketch of Frame and Panelled door? (plan , section & elevation) | (CO 3) | 6 |
| B) Draw a neat sketch of Louvered window? | (CO 3) | 6 |
| C) Draw a neat sketch of wooden stair showing all components? | (CO 3) | 6 |

Q.5 Solve Any One of the following.

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|--|--------|----|
| A) Explain king post truss and its joints with neat sketches? | (CO 3) | 12 |
| B) Explain Queen post truss and its joints with neat sketches? | (CO 3) | 12 |

Q.6 Solve Any two of the following.

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|---|--------|---|
| A) Explain merits and demerits of prefabrication? | (CO 4) | 6 |
| B) Describe various types of prefabrication? Explain the term 'Tolerance'? | (CO 4) | 6 |
| C) Describe various guidelines for transportation and erection in prefabrication? | (CO 4) | 6 |

*** End ***

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